SLAM performance predicition

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1 Introduction

In the field of autonomous mobile robotics, Simultaneous Localization And Mapping (SLAM) represents one of the most significant problems. As the name suggests, the goal of SLAM is incrementally obtain from a mobile robot a map of the environment and simultaneously locate the robot's position inside it. The difficulty of the SLAM problem is that, to precisely localize itself, the robot needs an accurate map, while to obtain an accurate map, it needs an accurate localization inside the map. This type of problems is called chicken-and-egg problem.

To get the map, SLAM algorithms take as input the observations generated by a laser sensor, typically Lidar, and the odometry data from the robot's motion sensors. The laser sensor provides distance measurements and angles to obstacles in the environment, allowing the robot to perceive its surroundings, while odometry supplies information about the robot's movement, such as changes in position and orientation over time. By combining these two sources of data, SLAM algorithms can incrementally build a map and estimate the robot's trajectory, even in previously unknown environments.

1.1 SLAM methods

In the last 30 years several approaches have been developed to solve the 2D SLAM problem leading to methods extensively used in both common and industrial scenarios. The main paradigms [1] used are:

- Extended Kalman Filter: a vector containing the estimates of the robot position and the environment landmark is used. In addition, a matrix composed of the correlations between the positions and landmark estimates is also used. Both the vector and the matrix are updated using the extended Kalman Filter.
- Particle Filter: a random sample is called a particle and it's composed of the robot position and a map estimate. Monte Carlo method is used to sample the possible states of the robot. The main steps are: predict the robot's pose for each particle, update particle weights based on sensor data, resample particles according to their weights, and update each particle's map.
- Graph-based: a graph is used where nodes represent robot poses or landmarks, and edges represent spatial constraints derived from sensor observations or odometry. The SLAM problem is formulated as a graph optimization task, where the goal is to find the configuration of nodes that best satisfies all constraints, typically using nonlinear optimization techniques.

1.2 Accuracy of SLAM methods

The heterogeneity of SLAM methods makes it difficult to find a measure that can evaluate their performance. Many approaches rely on manual evaluation or utilize information derived from the specific algorithm. Nowadays the most used metric is the Absolute Pose Error (APE)[2]. The APE measures the difference over time (Figure 1) between the estimated trajectory produced by the SLAM algorithm and the ground truth trajectory (Figure 2), focusing on the

global consistency of the estimated poses. The big advantage of the APE is that it allows for a quantitative assessment of how well the SLAM algorithm reconstructs the robot's trajectory over time, independent of the internal workings of the specific method used.

The APE is composed of two sub-metrics:

1. The Absolute Translational Error (ATE): it measures the difference in position, typically using the Euclidean distance between corresponding poses.
2. The Absolute Rotational Error (ARE): it measures the difference in orientation, often computed as the angular distance between corresponding rotations.

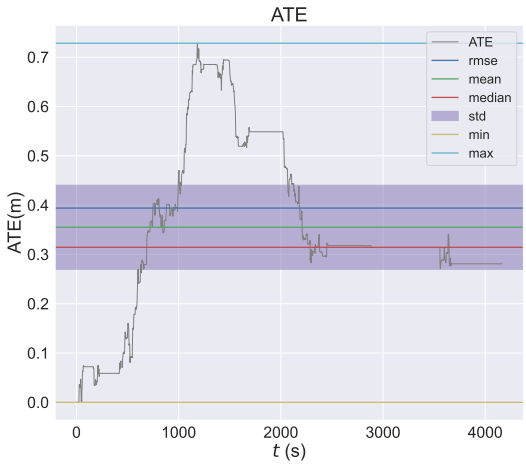

Figure 1: ATE variation over time

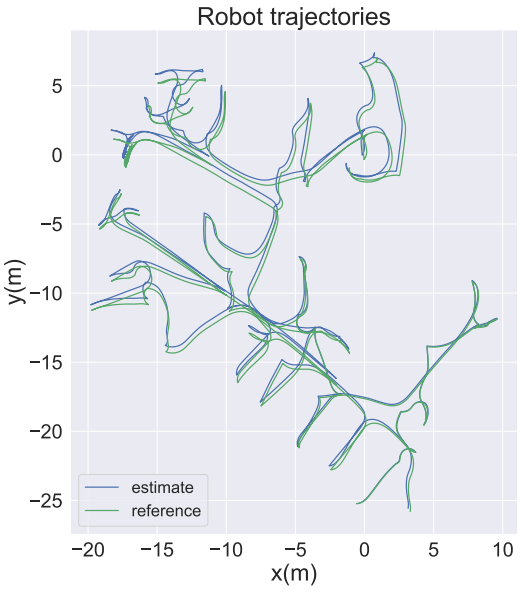

Figure 2: Difference between real (in green) and estimated (in blue) trajectory

1.3 APE calculation

As anticipated in the previous section, the APE metric is based on the difference between the estimated robot poses and the corresponding ground truth poses.

Formally, let $\boldsymbol{x}_{1:T}$ be the poses of the robot estimated by a SLAM algorithm from time step 1 to T, and let $\boldsymbol{x}^*_{1:T}$ be the corresponding ground truth poses.

Let $\delta_{i,j}$ be the relative transformation that moves the robot from pose x_i to x_j:

$$\delta_{i,j} = x_j \ominus x_i$$

$$\delta^*_{i,j} = x^*_j \ominus x^*_i$$

Let $trans(\cdot) \in \mathbb{R}^2$ be the function that extracts the translational (position) component from a transformation, and let $rot(\cdot) \in \mathbb{R}^2$ extract the rotational (orientation) component as Euler angle.

Finally, to compute the APE:

$$\text{APE} = \underbrace{\frac{1}{N}\sum_{i,j} trans(\delta_{i,j} \ominus \delta^*_{i,j})^2}_{\text{ATE}} + \underbrace{\frac{1}{N}\sum_{i,j} rot(\delta_{i,j} \ominus \delta^*_{i,j})^2}_{\text{ARE}}$$

Usually both APE and ATE and ARE are provided to assess SLAM performance.

1.4 Project goal

A major limitation of these metrics is their reliance on the availability of ground truth data, which is often difficult or impractical to obtain in real-world scenarios. Another aspect is that APE is calculated a posteriori, after the robot has explored the entire environment and so, these metrics can only be used to evaluate a run that that has already been completed.

The main goal of this project is to propose a predictor for the SLAM performance, specifically ATE and ARE, before running the SLAM algorithm or during its execution, using features and informations derived from a running robot and thus without any ground truth.

This type of prediction can improve the robustness and efficiency of autonomous mobile systems since the position error (APE) can be used to correct the robot trajectory.

The work in [3] proposes two main APE predictors, one based on linear regression, and another one based on a Gaussian Process via an RBF Kernel. Although Gaussian Process regression showed slightly better performance compared to linear regression model, the authors preferred linear regression because they considered it a more interpretable model. In both cases, a feature extractor is used [4] obtaining a set of structural features to represent the environment as a vector, which is then passed to the model.

In this work, three different predictors will be proposed, each based on popular machine learning CNN architectures.

2 Data

To train our models, a dataset derived from 309 robot exploring simulations will be used.

2.1 Dataset composition

The dataset is composed of 4613 datapoints. Each datapoint is represented as a tuple

$$(M, a, ate, are)$$

where:

1. $M \in [0, 255]^{n \times n}$: robot maps representend as grayscale maps that a robot has obtained during its navigation. Some examples are reported in Figure 3;
2. $a \in \mathbb{R}^+$: area measurements of the floorplan images, expressed in m^2;
3. $ate \in \mathbb{R}^+$: ATE values, expressed in m;
4. $are \in \mathbb{R}^+$: ARE values, expressed in rad;

Maps were obtained incrementally during all robot exploration, starting from the beginning and extracting the current slam robot map every 30 seconds wall time until the environment was fully explored. The choice to use incremental maps is due to the fact that we wants to estimate ATE and ARE online, while the robot is navigating.

The dataset has been randomly partitioned into training, validation, and test sets with proportions 70%, 15%, and 15% respectively.

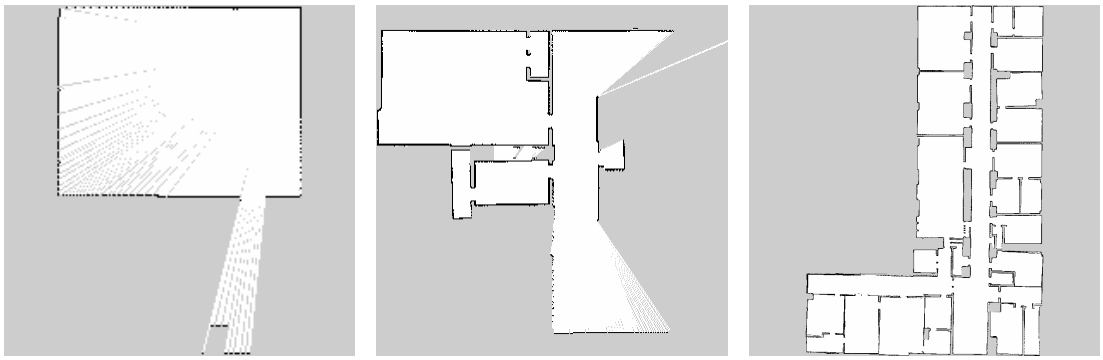

Figure 3: Examples of grayscale floorplan images used as input; the white pixels represent the free area, the black pixels the obstacles and the gray pixels the unknown area.

2.2 Data preprocessing

Evaluating the APE values shows that they are mainly distributed towards zero, with a significant presence of outliers, especially for ATE values. Figure 4 shows the histograms with highlighted the 0.99-quantile and the first value such that its Z-score is greater or equal to 3.5; the Z-score of a number x is $Z = \frac{x-\overline{x}}{\sigma}$ where $\overline{x}$ is the mean and σ the standard deviation.

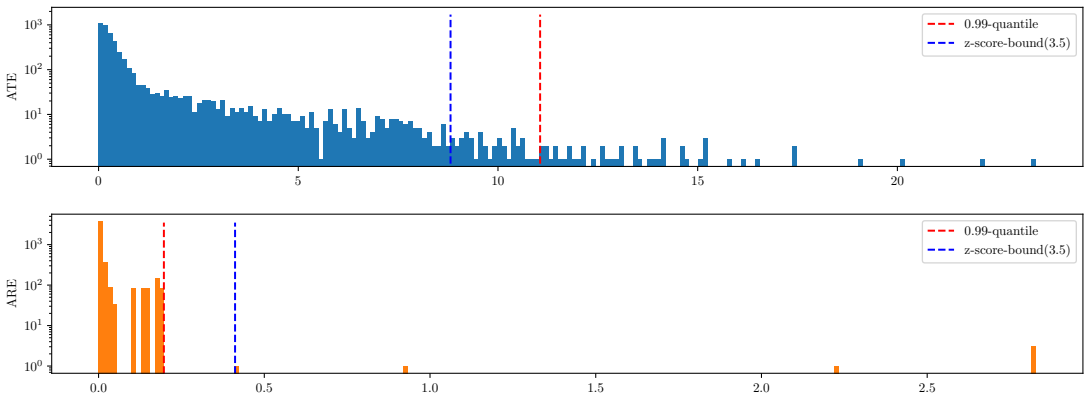

Figure 4: Box plot of the ATE (top) and ARE (bottom) values with log scale on the y axe; is also shown the 0.99-quantile (red line) and the first value with a z-score equal to 3.5 (blue line).

To facilitate our models to learn each quantity, the following strategies have been applied:

- Outliers cleaning: data points with ATE or ARE values above the 0.99-quantile or with a Z-score greater than or equal to 3.5 have been removed to reduce the impact of extreme outliers. 113 datapoints were removed;
- Logarithmic rescaling: due to the skewed distribution of ATE and ARE, a logarithmic transformation has been applied to these values to reduce the effect of large values and improve model learning; let x be the interested value to be rescaled, the applied operation is:
$$\log(1+x)$$
where log is the natural logarithm;
- Rescaling: after the logarithmic rescaling, a linear rescaling to $[0,1]$ has been applied; let $\{x_1,\dots,x_n\}$ be interested values to be rescaled, the applied operation is
$$\frac{x_i}{\max\limits_{1\le k\le n} x_k} \qquad \text{i=1,...,n}$$

Figure 5 shows data distribution after all the trasformations.

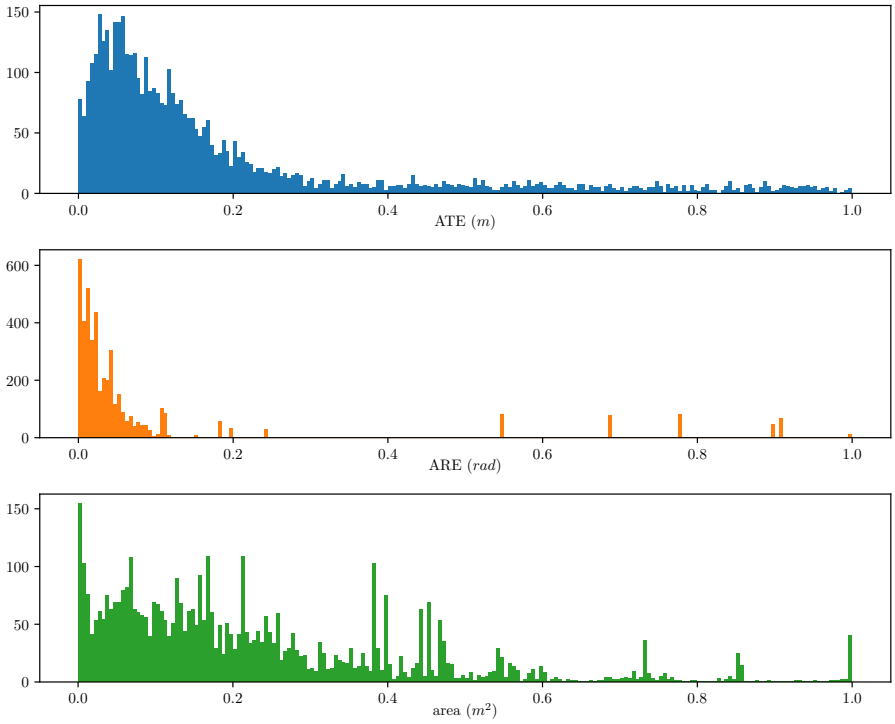

Figure 5: Dataset values after the log and linear rescaling.

- Image cropping: each map has been cropped to remove unnecessary borders, focusing only on the relevant area to reduce input size and eliminate irrelevant background; the crop is always squared in order to mantain this property to all floorplan images. Figure 6 shows an example of this operation;

Figure 6: A floorplan (left) and its bounding square crop (right).

- Image resizing: each floorplan image is resized to 224×224 pixels to ensure a uniform input size for the neural network models;

2.3 Data augmentation

Given the relatively limited size of the dataset, the following data augmentation techniques have been applied to training set:

- Random flip: both horizontal and vertical flips are applied;

- Random rotation: random rotation between $0°, 90°, 180°, 270°$ is applied;

3 Models

In this project, different architectures are used, all based on Convolutional Neural Networks (CNNs) to delegate feature extraction to the model and an head fully connected network (FC) to map these features, together with the map area, to the target regression outputs (ATE and ARE). The map area value skip the CNN because, otherwise, the scaling done by the model in the initial stage would cause the true value of the area to be lost.

ReLU function, defined as $\text{ReLU}(x) = \max(0, x)$, is applied to the output of each models; this to ensure that the predicted ATE and ARE are non negative values.

To reduce overfitting, dropout technique [5] is applied within the fully connected layers of each model. Lastly, Batch Normalization technique [6] is applied after the convolutional layers of the models in order to accelerate the convergence and reduce overfitting.

Three distinct models were used:

1. AlexNet [7], one of the first successful architectures in the field of CNNs. It will be a baseline for the other models;
2. MobileNetV3 [8], the state of the art of lightweight CNN architectures designed for mobile and embedded vision applications;
3. SqueezeNet [9], an ultra-compact CNN with few parameters, offering decent accuracy for low-memory device;

The choice of the last two architectures is aimed at achieving a lightweight predictor that can easily run on limited hardware, such as mobile robots or embedded systems. These models, indeed, are designed to minimize memory usage and computational requirements, making them suitable for real-time applications where resources are constrained.

Formally, the model h we aim to learn can be expressed as follows:

$$h : [0, 255]^{n \times n} \times \mathbb{R}^+ \to \mathbb{R}^2$$

where $[0, 255]^{n \times n}$ is a robot map encoded as an $n \times n$ grayscale image, $\mathbb{R}^+$ is the area of the floorplan expressed in m^2, and $\mathbb{R}^2$ contains ATE (m) and ARE (rad).

3.1 AlexNet

To begin, a model based on AlexNet was chosen, one of the first successful architectures in the field of convolutional neural networks. AlexNet consists of several convolutional layers followed by fully connected layers, using ReLU activations and max-pooling. The relatively large number of layers without the use of modern techniques to lighten its computational load, makes this model a deep CNN resulting in a high number of parameters ($\approx 37\,298\,882$). The abstract structure is shown on Figure 7.

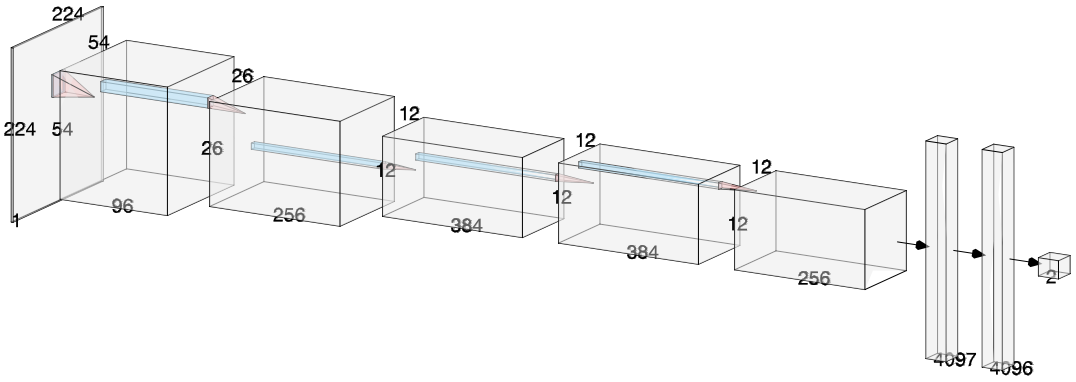

Figure 7: Model structure based on AlexNet. Realized with [10].

Two main sections can be outlined:

1. A CNN feature extractor: it processes the input image through a series of convolutional and max pooling layers, resulting in a condensed set of features;

2. A Regression head: it predicts ATE and ARE values through some fully connected layers whose input is the set of features extracted from the CNN;

ReLU is used as the activation function in both the convolutional and fully connected layers.

3.1.1 Batch Normalization

A batch normalization layer was added between each convolutional layer, before ReLU. This layer normalizes the output to have zero mean and unit variance applying the following trasformation:

$$\text{BatchNorm}(x) = \hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}}$$

and then translates and scales:

$$y = \gamma \hat{x} + \beta$$

where γ and β are two learned parameters, μ_B is the mean of the batch B, σ_B^2 is the variance of the batch B and ε is an arbitrary small constant added for numerical stability.

3.1.2 Dropout

Dropout technique is applied within the fully connected layers of the regression head. Dropout randomly sets a fraction of the input units to zero during training, which prevents the network from relying too heavily on specific features.

3.2 MobileNetV3

Computational requirements that AlexNet needs make it unsuitable for hardware used in mobile robotics. For this reason, MobileNetV3 was selected as a more efficient alternative, offering a state-of-the-art trade-off between accuracy and computational efficiency. In particular, the MobileNetV3 small version was used as starting point, which has ≈1 520 642 parameters, 24 times less than AlexNet.

To achieve this result, different optimization techniques are used, but we will focus only on the main ones.

3.2.1 Depthwise and Pointwise Convolutions

Convolutional layers on which CNNs are based have led to massive achievements in machine learning projects, especially in computer vision. However, standard convolutions are computationally expensive, particularly for deep networks. MobileNetV1 addresses this by employing depthwise separable convolutions, which split the convolution operation into two parts:

1. Depthwise convolution: processes each input channel separately with its own filter, extracting spatial features independently for each channel;
2. Pointwise convolution: applies 1×1 convolutions across all channels, combining the outputs from the depthwise step to integrate information across channels;

Figure 8 shows the difference between standard convolution and depthwise convolution.

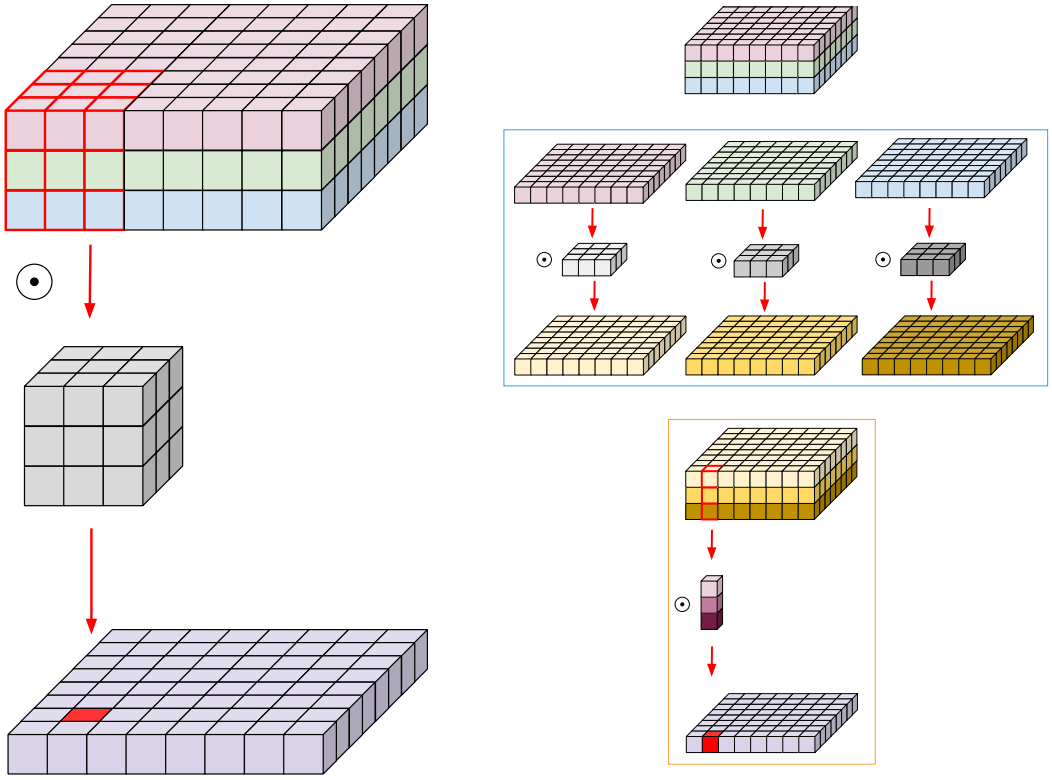

Figure 8: Standard Convolution (left) and Deepthwise+Pointwise Convolution (right). The Deepthwise conv. is highlighted in light blue while Pointwise conv. in orange [11].

To understand computational savings in terms of number of multiplications, take the follwing example: let S be the dimensions of the squared input, K the dimensions of the squared kernel filter, C_{in} the number of input channels and C_{out} the numbers of output channels. We'll assume:

$$S = 128 \quad K = 3 \quad C_{in} = 3 \quad C_{out} = 16$$

	Regular convolution	Deepthwise and Pointwise convolution
Parameters	$3^2 \cdot 3 \cdot 16 = 432$	$3^2 \cdot 3 \cdot 128^2 \cdot 16 = 7\,077\,888$
# multiplications	$3^2 \cdot 3 + 3 \cdot 16 = 75$	$3^2 \cdot 3 \cdot 128^2 + 128^2 \cdot 3 \cdot 16 = 1\,228\,800$

3.2.2 Squeeze and Excitation

Introduced in [12], Squeeze and Excitation is a technique that focuses on the interdependencies between channels. As the name suggests, there are two phases:

1. Squeeze phase: each channel is squeezed into a single value through the use of averge pooling. The output of this phase is then a vector $\boldsymbol{z}$ of C elements, where C is the number of channels:

$$\boldsymbol{z} = \{z_1, \ldots, z_C\}$$

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} u_c(i,j) \qquad \text{c=1,...,C}$$

where $\boldsymbol{U} = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_C\}$ is the set of feature maps of size $H \times W$ and $u_c(i,j)$ is the element in position (i,j) on the c-th feature map.

2. Excitation phase: the squeezed vector $\boldsymbol{z}$ is fed into a two-layer feed-forward neural network composed of a first layer of size C/r with ReLU and a second layer of size C with sigmoid, where $\text{sigmoid}(x) = 1/(1+e^{-x})$ and r is a reduction ratio. The purpose of this phase is to capture the intricate dependencies among channels. The output of this network is another vector $\boldsymbol{s}$ of the same dimensions of $\boldsymbol{z}$, containing the importance weights for each channel.

3. Scale and combine phase: the importance weights vector $\boldsymbol{s}$ is used to rescale each channel of the original input by channel-wise multiplication ($*$):

$$\tilde{\boldsymbol{x}}_c = s_c * \boldsymbol{u}_c \qquad \text{c=1,...,C}$$

where $\tilde{\boldsymbol{X}} = \{\tilde{\boldsymbol{x}}_1, \ldots, \tilde{\boldsymbol{x}}_C\}$ is the final output. In this way most important feature are highlighted and the less important ones are downplayed.

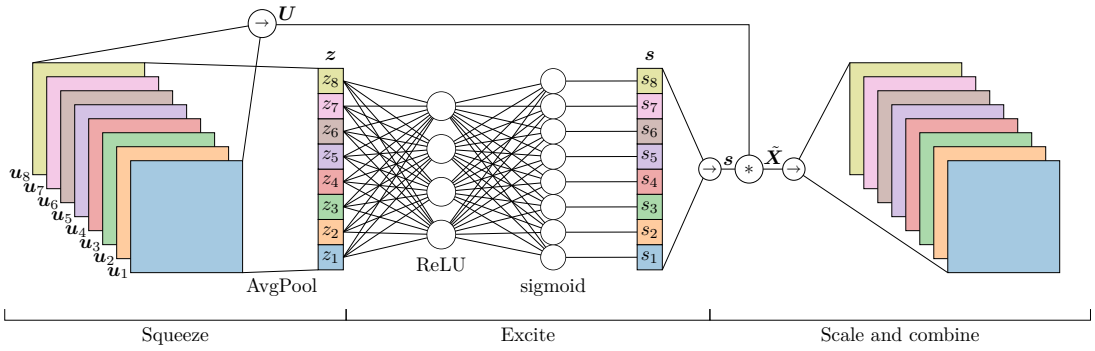

Figure 9: A graphical representation of Squeeze and excitation; in this case a reduction ratio of $r = 2$ is used with 8-channel feature maps.

3.2.3 HardSwish Activation function

To improve the accuracy of the neural networks ReLU was replaced by a new activation function called *swish*, defined as:

$$\text{swish}(x) = x \cdot \sigma(x)$$

where σ is the sigmoid function $\sigma(x) = 1/(1+e^{-x})$. However, despite swish improves accuracy, it is computationally expensive for mobile devices. To address this, MobileNetV3 introduces the *HardSwish* activation function, which approximates swish while being more efficient to compute. HardSwish is defined as:

$$\text{h-swish}(x) = x \cdot \frac{\text{ReLU6}(x+3)}{6}$$

where $\text{ReLU6}(x) = \min(\max(0,x), 6)$. This function retains the benefits of swish but is faster and more suitable for deployment on resource-constrained hardware.

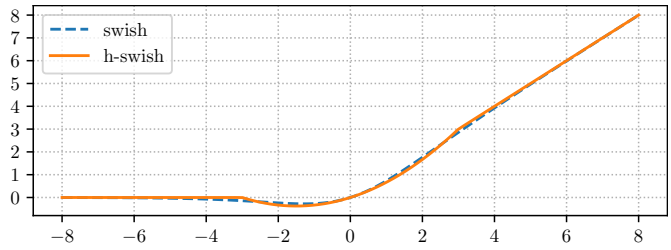

Figure 10: Swish and h-swish

3.3 SqueezeNet

As final model SqueezeNet was choosen. SqueezeNet is an ultra-compact convolutional neural network architecture designed to achieve AlexNet-level accuracy with significantly fewer parameters, as the original paper title "AlexNet-level accuracy with 50x fewer parameters and <0.5MB model size" [9] suggests; as proof of this, the model that was used, derived from a minor modification of SqueezeNet v1.1, has ≈722 740 params, ∼51.6 times less than AlexNet models described in section 3.1. The few parameters of SqueezeNet also led to a large reduction in power consumption and inference time, making it one of the most widely used architectures in various fields such as autonomous driving, IoT, and mobile AI.

A batch normalization layer was added after every fire module (explained in section 3.3.1) and convolutional layer.

Another aspect of SqueezeNet is the absence of a Fully Connected (FC) layer; the authors made this choice because these kind of layers use a large number of parameters. In our model, shown in Figure 11, a small MultiLayerPerceptron (MLP) was added so that the area value was also utilized. The MLP is composed of 3 linear layers with ReLU as activation function and dropout. This addition resulted in a slight increase in the model parameters, precisely 35010.

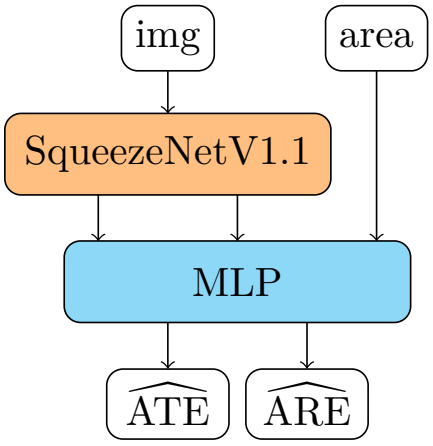

Figure 11: Model structure based on SqueezeNet v1.1

To understand how SqueezeNet was able to reduce the number of parameters consider a convolutional layer formed only with 3×3 filters. Its numbers of parameters is:

$$(\#\text{ filters}) \cdot (\#\text{ input channels}) \cdot (3 \cdot 3)$$

So, to maintain a small number of parameters in a CNN it is important to reduce both the number of filters and the number of input channels. The authors of [9], following this reasoning, have employed three strategies:

1. Replace most of 3×3 filters with 1×1 filter: in this way they reduce the numbers of 3×3 filters by replacing them with 1×1 filters that have 9 times fewer parameters;

2. Decrease the number of input channels of 3×3 filters: to do so squeeze layers was used (similar but different from the squeeze phase described in secction 3.2.2);

3. Downsaple late in the network: in this way convolutional layers have large activation map leading to higher accuracy [13];

To implement strategy 3, max-pooling with a stride of 2 is performed after four layers, while, for the first two strategies, Fire modules are introduced.

3.3.1 Fire modules

A Fire module consists of:

- A squeeze convolution layer: it uses $s_{1\times1}$ filters 1×1 to squeeze all the input channels into $s_{1\times1}$ single channel feature maps (strategy 1); shown in Figure 12;
- An expand layer: formed by $e_{1\times1}$ filters 1×1 and $e_{3\times3}$ filters 3×3; it takes as input the $s_{1\times1}$ feature maps and returns the $e_{1\times1} + e_{3\times3}$ feature maps obtained from the convolutional filters of this layer. To limit the number of input channels of the 3×3 filters, Fire modules set $s_{1\times1}$ to be less than $e_{1\times1} + e_{3\times3}$ (strategy 2).

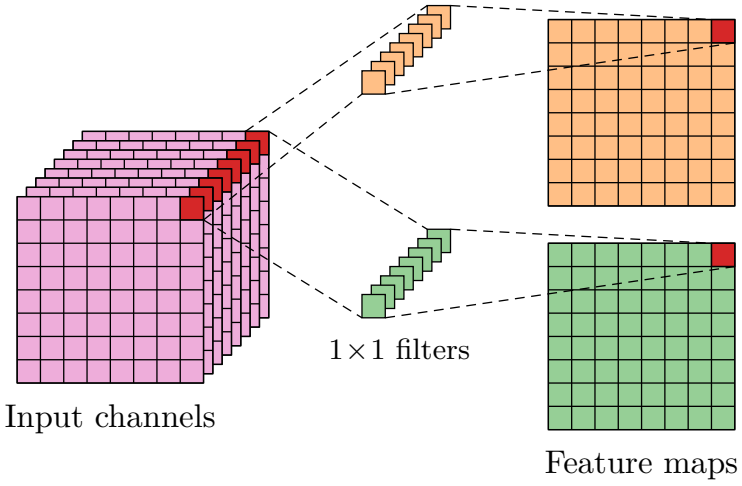

Figure 12: Squeeze convolutional layer with $s_{1\times1} = 2$ and 8 input channels.

By stacking multiple fire modules, SqueezeNet achieves a significant reduction in the number of parameters while maintaining competitive accuracy.

4 Hyperparameter optimization

Now that the models have been defined in the previous section, an Hyperparameter (HP) optimization phase is needed. The HP choice is essential to obtain the best performance of a neural network, and it can significantly influence the model's ability to generalize to unseen data.

Only 4 Hyperparameters [with related possible values] are considered in the models:

1. Learning rate: the step size at each iteration while moving toward a minimum of the loss function; it controls how much the model weights are updated during training. $[0.0001, 0.01]$

2. Batch Size: the number of samples used to compute each update of the model weights; it influences the stability and speed of the training process. $\{16, 32, 64, 128\}$

3. Dropout percentage: the fraction of input units to drop during training to prevent overfitting; it helps improve the model's ability to generalize. $[0, 0.5]$

4. Momentum: coefficient that accelerates optimization by accumulating past gradients; helps achieve faster and more stable convergence. $[0, 1]$

Given the use of state-of-the-art architectures, other HPs such as activation functions, layers number, and layer types have not been changed.

4.1 Search Algorithms

The search for optimal HP values can be done in two ways: manually setting them based on prior experience or intuition, or using automated search algorithms.

Although manual testing remains one of the most widely used methods, especially in the student environment, it requires a deep knowledge of the used algorithms and their HPs meaining. In addition, it has been found that a manual procedure is ineffective for several problems due to different factors, such as the high dimensionality of the hyperparameter space, the presence of complex interactions between hyperparameters, and the computational cost of exhaustively evaluating all possible combinations [14]. For this reasons, manual testing was discarded in favor of using a search algorithm.

Formally, a HPs optimization problem aims to find the HPs configuration $\boldsymbol{x}^* \in X$ such that:

$$\boldsymbol{x}^* = \underset{\boldsymbol{x} \in X}{\operatorname{argmin}} f(\boldsymbol{x})$$

where X is the search space containing all the possible hyperparameter configurations, and f is the objective function (e.g., validation loss or error) to be minimized. A search algorithm is an heuristic algorithm whose goal is to achieve the optimal, or near-optimal, HPs configuration within a given budget, which can be of time or computational resources.

In the next section the simplest search algorithms will be shown. Given the low number of HPs, the use of more sophisticated algorithms like Bayesan optimization or Gradient based optimization was not deemed necessary in favor of a simpler random search.

4.1.1 Grid search

Grid search is the most straightforward hyperparameter tuning algorithm. It discretizes the search space of each hyperparameter and evaluates all possible combinations. The combination that yields the best average performance is selected as the optimal set of hyperparameters.

The well-performance regions cannot be detected, however, to identify the global optimums, the following procedure can be performed:

1. Start with a large search space and step size;

2. Narrow the search space and step size based on the best performance regions found on the previous step;

3. Repeat step 2 multiple times until an optimum is reached.

Grid search can be easily be implemented and parallelized but, being de facto brute force, it's ineffective with a high number of HPs. In fact, its computational complexity is $O(n^k)$ where k is the HPs number and n is the number of possible values.

4.1.2 Random search

Random search is a search algorithm similar to grid search. Instead of exploring all the values in the search space, random search randomly samples hyperparameter combinations from the defined ranges or sets. Each sampled configuration is evaluated, and the best-performing set is selected.

With a limited budget, random search is able to explore a larger search space than grid search by reducing the probability of wasting time on small poor-performance regions. Its

computational complexity is $O(n)$, where n is the number of sampled configurations defined before the optimization process starts.

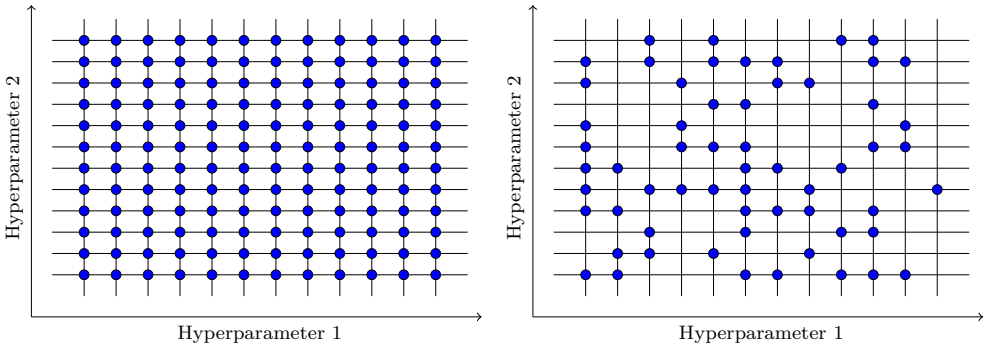

Figure 13: Comparison between grid search(left) and random search(right) with 2 HPs

The random search and grid space problem is that every evaluation is independent of previous evaluations; thus, they waste massive time evaluating poorly-performance areas of the search space.

4.2 Setup

To perform the HPs optimization phase Ray Tune [15], an industrial standard tool for distributed HP tuning, was used. To find the optimal HPs configuration a random space was performed with the following search spaces:

1. Learning rate: the search was performed using `tune.loguniform` between $[0.0001, 0.01]$, which samples values x such that $\log x$ is uniformly distributed between $\log a$ and $\log b$;
2. Batch size: $\{16, 32, 64, 128\}$;
3. Dropout percentage: $[0, 0.6]$ divided in 10 equally spaced points.
4. Momentum: $[0, 1]$ divided in 20 equally spaced points.

To run every HPs configuration trial an Asynchronous Successive Halving Algorithm (ASHA) scheduler [16] was used.

4.2.1 Asynchronous Successive Halving Algorithm

ASHA is an early stopping algorithm that efficiently allocates resources by quickly terminating underperforming trials and allocating more resources to promising ones.

More precisely, ASHA's procedure can be summarized as follows:

1. Launch many trials, one per HPs configuration;
2. Every trial is performed for at least g time units;
3. After g time units all trials that have reached the same amount of time units are evaluated;
4. Only a $1/r$ fraction of the best trials are kept, while the others are early stopped;
5. The remaining trials continue to be performed and will be evaluated in the next round;
6. Only the trials that survive all rounds are trained until the maximum number of time units or until convergence.

where g is the minimum number of time units that each trial must run before being considered for early stopping, and r is the reduction factor determining the proportion of trials retained at each round. Time units can be any resource such as epochs, fold of CV, or wall-clock time, depending on the specific application.

This approach significantly reduces the computational cost compared to evaluating all random configurations for the full number of time units, while still identifying high-performing hyperparameter settings.

4.3 Results

For each model, 20 random hyperparameter configurations were analyzed using 5-fold cross-validation to evaluate their performance. For each fold 10 epochs were performed. CV folds were used as time units with a minimum of 2 folds (g) and a maximum of 5 folds. In this way, if a configuration performed poorly on the first 2 folds it was stopped early by the ASHA scheduler as it is unlikely that it would outperform other configurations.

The results of HPs optimization are shown in Table 1.

Hyperparameter	**AlexNet**	**MobileNetV3**	**SqueezeNet**
Learning rate	1.2565×10^{-4}	6.3534×10^{-4}	3.6475×10^{-3}
Batch size	32	32	32
Dropout perc.	0.2	0.3	0.1
Momentum.	0.9	0.85	0.85

Table 1: HPs optimization results

7.5 hourse were required to achieve these results with an NVIDIA RTX 3090 as GPU and an Intel Core i7-12700K as CPU.

5 Training

In this section, the training phase of the model is described, followed by an evaluation of performance on real robot navigations.

Mean Squared Error (MSE) was chosen as loss function; other functions like MAE or Huber loss were tested but showed lower performance than MSE. The training duration was 50 epochs. $k-$fold Cross Validation ($k-$CV) was used to estimate the generalization performance and to mitigate overfitting. For each fold, the model was trained on $k-1$ subsets, always with 50 epochs, and validated on the remaining one, rotating through all folds. This approach ensures that each data point is used for both training and validation, providing a more robust evaluation of model performance. A k value of 5 was used.

Each model was trained from scratch through the Stochastic Gradient Descent (SGD) optimizer with learning rate and momentum found in the previous section and using batch size and dropout found in the previous section.

5.1 Training results

5.1.1 Prediction performance

Performance results are shown in Table 2 in terms of losses and Table 3 in terms of R^2-Score, RMSE and NRMSE. Looking at the first table, SqueezeNet outperforms the others in all losses except for the $5-$CV loss, suggesting a worse generalization of the model. AlexNet and MobileNetV3 show similar results in all losses with a slight advantage of the latter and a good generalization.

Models	Train loss	Valid. Loss	Test Loss	5−CV Loss
AlexNet	0.01804	0.01808	0.01691	0.01718
MobileNetV3	0.01615	0.01807	0.01718	0.01667
SqueezeNet	0.01340	0.01437	0.01309	0.02040

Table 2: Models performances in terms of loss values

In Table 3 the following metrics are used:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2} \qquad \text{RMSE} = \sqrt{\frac{\sum_i (y_i - \hat{y}_i)^2}{n}} \qquad \text{NRMSE} = \frac{\text{RMSE}}{\max_i \hat{y}_i - \min_i \hat{y}_i}$$

where y_i is the i-th real value and $\hat{y}_i$ its prediction, $\overline{y}$ the mean of the real values and n the number of data points.

Here again SqueezeNet outperforms the other models in all cases except NRMSE for ATE where it still gets a good value compared to the others. AlexNet has Pareto worst performance while MobileNetV3 stays in the middle.

	ATE			ARE		
Models	R^2	**RMSE**	**NRMSE**	R^2	**RMSE**	**NRMSE**
Alexnet	0.4	1.19	0.28	0.81	0.017	0.086
MobileNetV3	0.46	1.13	0.164	0.85	0.016	0.078
SqueezeNet	0.5	1.09	0.19	0.92	0.011	0.056

Table 3: Performance comparison in terms of R^2-Score, RMSE (m) and NRMSE (% with respect to the range of predicted values)

The learning curves obtained during the training phase are shown in Figure 14. The 5−CV loss and the test loss are highlighted, providing a visual comparison of the models generalization capabilities and their performance on unseen data.

In all models no evident signs of overfitting are present as the validation loss never deviates too much from the training loss. AlexNet has the smoothest descent. MobileNetV3 in the first 25 epochs shows a small gap between validation and training loss which, however, is narrowing in later epochs. In both AlexNet and MobileNetV3 test and 5−CV loss are very similar and close to the learning curves indicating good generalization of the models as the models perform well on both seen and unseen data. SqueezeNet shows a similar situation with a higher 5−CV indicating a possible lower capacity of the model to generalize.

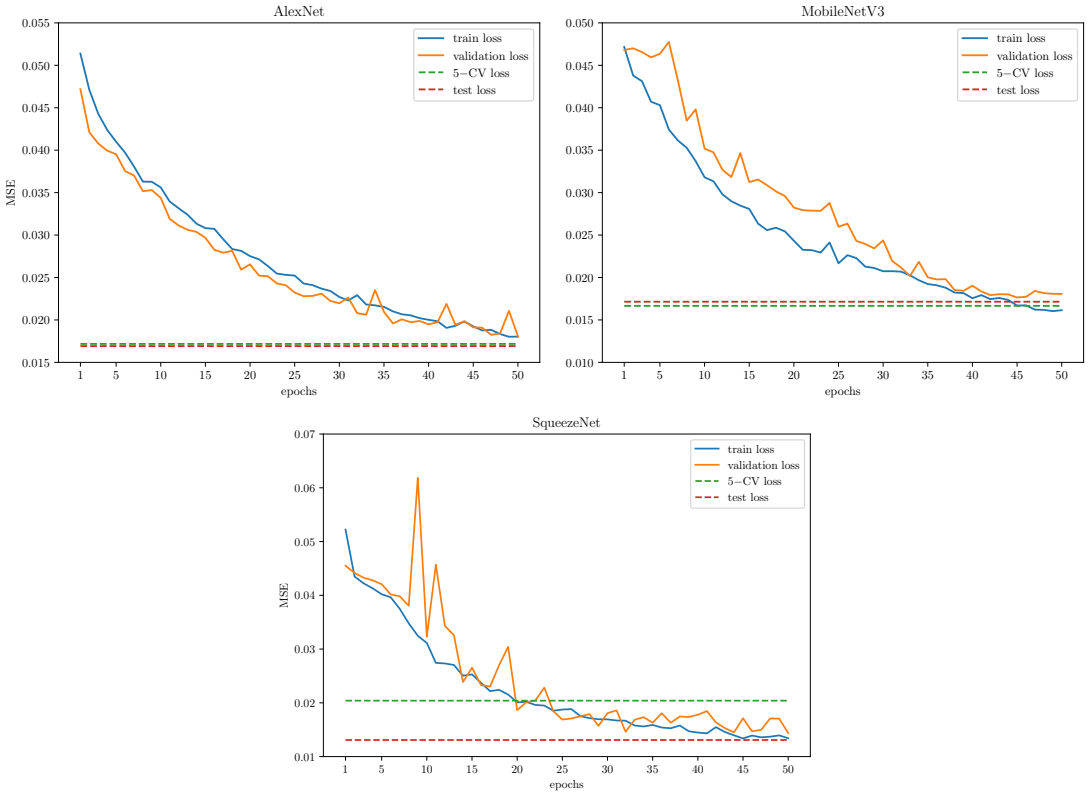

Figure 14: Learning curves of models; highlighted also the 5−CV loss and the test loss

5.1.2 Efficency

Regarding efficiency, in terms of inference time and model size, shown in Table 4, as expected, SqueezeNet is the most lightweight model and AlexNet, by far, the heaviest.

MobileNetV3 turns out to be, in general, the fastest model with the lowest CPU time and a slightly slower GPU than AlexNet, which, counterintuitively, is the fastest in terms of GPU time. The latter fact is mainly due to the simple structure of the model, consisting of only basic operations that are highly optimized in modern GPU.

So, overall, MobileNetV3 finishes as the most efficient model among those seen in this work.

Inference times were calculated by averaging the individual datapoint inference times over the test set. Only batchs of one element are used. This because we wanted to analyze inference times in real time applications, where what matters is immediate responsiveness and not throughput.

Models	tCPU(ms)	tGPU(ms)	Size(MB)
AlexNet	16.8166	0.2523	142.3
MobileNetV3	1.4469	0.5893	5.86
SqueezeNet	3.1651	0.3553	2.76

Table 4: Models inference times with CPU and GPU and sizes

5.2 SLAM predicition evaluation

It should be highlighted that all models have a very high RMSE for the ATE, as Table 3 shows. An error of 1 m, when used to correct SLAM prediction of trajectories, could be too high and would irreparably damage the robot's navigation. To understand this problem, a small analysis was conducted.

5.2.1 Analysis of high RMSE

Let $R = \{r_1, \ldots, r_n\}$ be the set of real ATE values present in dataset and $P = \{p_1, \ldots, p_n\}$ be the corresponding predicition. Figure 15 shows the scatter plots of each model with the real values in the x axis and the error $r_i - p_i$ between the real values and predictions in the y axis. The data used are from the test set.

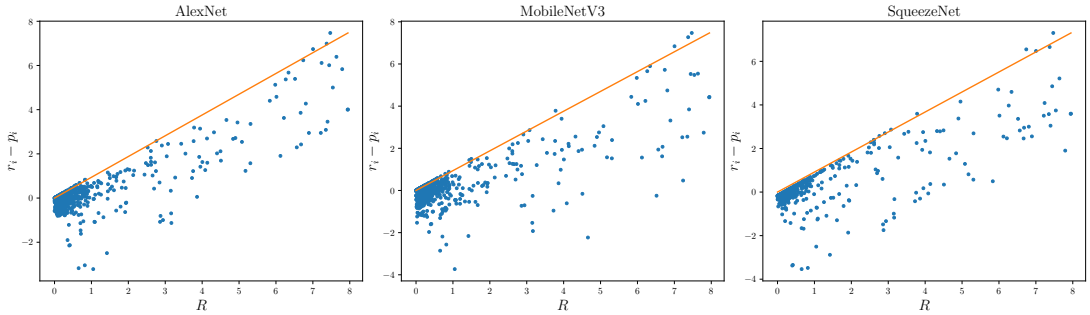

Figure 15: Scatter plots for real values and absolute prediction error. In orange the bisector of the positive axes

It is evident from the figures a correlation between the real values and the prediction error: the higher the real ATE value, the more the predictor is wrong. This simply means that most of the high ATE values are predicted with a low values and thus the error remains high. In this way the models failed to learn with big ATE values leading to having such a high RMSE. The very skew distribution, showed in Figure 16, combined with the low dataset dimension, could be the main causes of this.

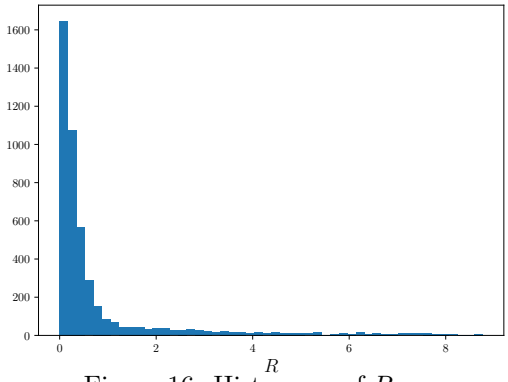

Figure 16: Histogram of R

5.2.2 Incremental APE evaluation

6 Conclusions

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